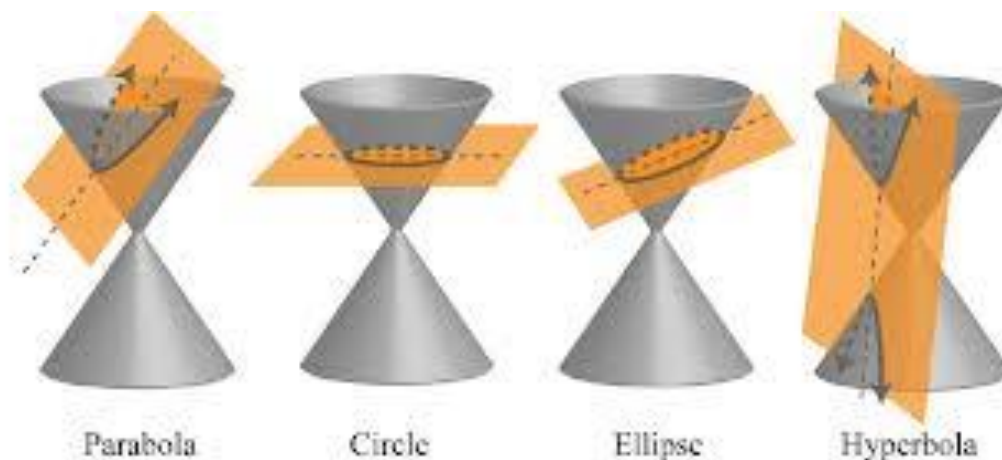


5.4.1 Conic Sections (Parabolas)

If a plane passes through a double right circular cone there are four possible resulting curves. These curves are the **parabola**, **circle**, **ellipse**, and **hyperbola**. These curves are called **conic sections** and are shown below.



Definition: A **parabola** is the set of all points in the plane that are equidistant from a fixed line (the **directrix**) and a fixed point not on the line (the **focus**)

The Equation of a Parabola: (opening up or down)

The parabola

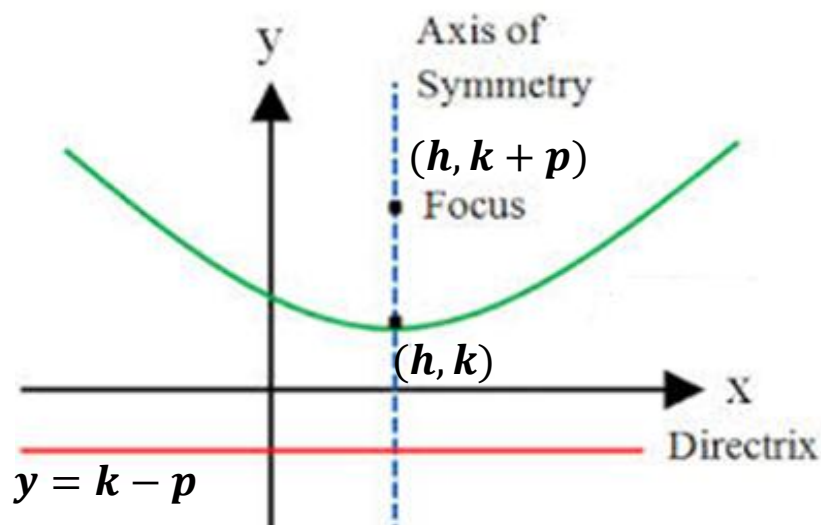
$$y = a(x - h)^2 + k$$

has focus $(h, k + p)$

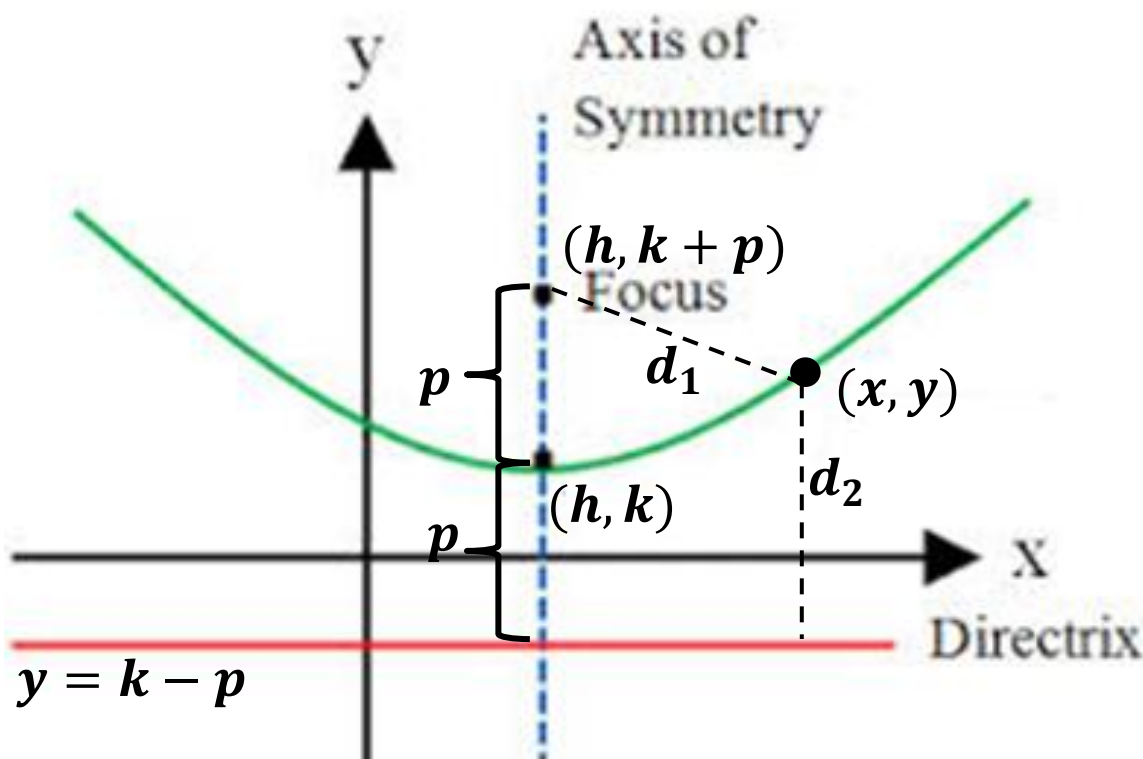
and directrix $y = k - p$

where $a = \frac{1}{4p}$ and (h, k) is the vertex.

Note: If $a > 0$ then the parabola opens up. If $a < 0$ then the parabola opens down.



Developing the Equation of a Parabola



We see that $d_1 = d_2$

So

$$\begin{aligned}\sqrt{(x-h)^2 + (y-(k+p))^2} &= \sqrt{(x-x)^2 + (y-(k-p))^2} \\ \Leftrightarrow (x-h)^2 + (y-(k+p))^2 &= 0 + (y-(k-p))^2 \\ \Leftrightarrow (x-h)^2 + y^2 - 2y(k+p) + (k+p)^2 &= y^2 - 2y(k-p) + (k-p)^2 \\ \Leftrightarrow (x-h)^2 + y^2 - 2yk - 2yp + k^2 + 2kp + p^2 &= y^2 - 2yk + 2yp + k^2 - 2kp + p^2 \\ \Leftrightarrow (x-h)^2 - 2yp + 2kp &= +2yp - 2kp \\ \Leftrightarrow (x-h)^2 + 4kp &= 4yp \\ \Leftrightarrow \frac{1}{4p}(x-h)^2 + k &= y\end{aligned}$$

Example 1: Find the equation of the parabola with focus $(-1, 3)$ and directrix $y = 2$

Since the directrix is a horizontal line we know this is parabola that opens up or down.

Focus: $(h, k + p)$ so $h = -1$ & $k + p = 3$

Directrix: $y = k - p$ so $k - p = 2$

We solve the system $\begin{cases} k + p = 3 \\ k - p = 2 \end{cases}$ and find, $p = \frac{1}{2}$ & $k = \frac{5}{2}$.

Therefore our equation is:

$$y = \frac{1}{4p}(x - h)^2 + k \rightarrow y = \frac{1}{2}(x + 1)^2 + \frac{5}{2}$$

Example 2: Find the vertex, focus, and directrix of the graph of $y = -3x^2 - 6x + 2$.

Since this equations shows y as a function of x we know this is a parabola that opens up or down. Let's first get our quadratic into vertex form by completing the square:

$$y = -3(x^2 + 2x + \quad) + 2 \Leftrightarrow y = -3(x^2 + 2x + 1) + 2 + 3 \Leftrightarrow y = -3(x + 1)^2 + 5$$

$$\text{So } \frac{1}{4p} = 3 \Leftrightarrow p = -\frac{1}{12}, \quad h = -1, \quad k = 5$$

$$\text{Vertex: } (h, k) \rightarrow (-1, 5) \quad \text{Focus: } (h, k + p) \rightarrow \left(-1, \frac{59}{12}\right)$$

$$\text{Directrix: } y = k - p \rightarrow y = \frac{61}{12}$$

Example 3: Find the vertex, focus, and directrix of the graph of $y = 2x^2 + 6x - 7$.

Since this equations shows y as a function of x we know this is a parabola that opens up or down. Let's first get our quadratic into vertex form by completing the square:

$$y = 2(x^2 + 3x + \quad) - 7 \Leftrightarrow y = 2\left(x^2 + 3x + \frac{9}{4}\right) - 7 - \frac{9}{2} \Leftrightarrow y = 2\left(x + \frac{3}{2}\right)^2 - \frac{23}{2}$$

$$\text{So } \frac{1}{4p} = 2 \Leftrightarrow p = \frac{1}{8}, \quad h = -\frac{3}{2}, \quad k = -\frac{23}{2}$$

$$\text{Vertex: } (h, k) \rightarrow \left(-\frac{3}{2}, -\frac{23}{2}\right) \quad \text{Focus: } (h, k + p) \rightarrow \left(-\frac{3}{2}, -\frac{91}{8}\right)$$

$$\text{Directrix: } y = k - p \rightarrow y = -\frac{93}{8}$$

The Equation of a Parabola: (opening left or right)

The parabola

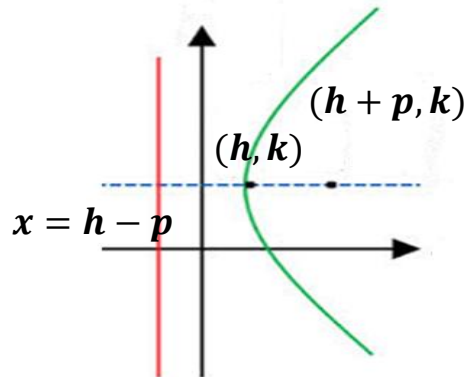
$$x = a(y - k)^2 + h$$

has focus $(h + p, k)$

and directrix $x = h - p$

where $a = \frac{1}{4p}$ and (h, k) is the vertex.

Note: If $a > 0$ then the parabola opens to the right. If $a < 0$ then the parabola opens to the left.



Example 4: Find the vertex, axis of symmetry, y-intercepts, focus, and directrix for the parabola $x = y^2 - 2y$. Sketch a graph of this parabola.

Since this equation shows x as a function of y we know this is a parabola that opens left or right. Let's first get our quadratic into vertex form by completing the square:

$$x = y^2 - 2y \Leftrightarrow x = y^2 - 2y + 1 - 1 \Leftrightarrow x = (y - 1)^2 - 1$$

$$\text{So } \frac{1}{4p} = 1 \Leftrightarrow p = \frac{1}{4}, \quad h = -1, \quad k = 1$$

$$\text{Vertex: } (h, k) \rightarrow (-1, 1) \quad \text{Focus: } (h + p, k) \rightarrow \left(-\frac{3}{4}, 1\right)$$

$$\text{Directrix: } x = h - p \rightarrow x = -\frac{5}{4} \quad \text{Axis of Symmetry: } y = k \rightarrow y = 1$$

y-intercept(s) will occur when $x = 0$.

$$0 = y^2 - 2y$$

$$0 = y(y - 2)$$

$$y = 0 \text{ or } y = 2$$

y-intercepts: $(0, 0)$ & $(0, 2)$

